I) Def: A metric space is a set X with a metric d.

Denote (X, d) where X is the set and d is the metric

Examples

- $(\mathbb{R}^2, d_2)$

- $\ell = \{$set of sequences $(x_0, x_1, \dots)$ s.t. $\sum_{i=0}^{\infty} |x_i|^p < \infty$

  (ℓ, d_p) : metric space

- $\ell_\infty = \{$set of sequences $(x_0, x_1, \dots)$ s.t $|x_n| < \infty \ \forall n\}$

  (ℓ_∞, d_∞) : Metric space

- $X = C[a,b]$ = set of continuous functions on set $[a,b]$

  $$d_p(x,y) = \left[\int_a^b |x(t) - y(t)|^p \, dt\right]^{1/p} \Bigg\} \text{ LP Metric}$$

- $X = C[a,b]$

  $$d_\infty(x,y) = \sup_{t \in [a,b]} \left(|x(t) - y(t)|\right)$$

$$x_0(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t} \quad \text{(Fourier Series)}$$

II Vector Space

Def: A vector space S over a field of scalars R is a collection of objects known as vectors, together with an addition operation, $+$, and scalar multiplication operation, $\cdot$, that satisfy the following

1) S forms an abelian group under addition

a) If $x \in S$ and $y \in S$ $x+y \in S$

(closure under addition)

b) There is an element, $0 \in S$, s.t. $x+0 = x$ and $0+x = x$

(0 is the additive identity)

c) For every element $x \in S$ there is a $y \in S$ s.t.

$x+y=0$
$y+x=0$ (y is the additive inverse of x)

d) Addition commutes: $x+y = y+x$ (Abelian)

e) Addition is associative: $a, b, c \in S$

$(a+b)+c = a+(b+c)$

2) For any $a, b \in R$ and $x, y \in S$

$ax \in S$ (closure in scalar multiplication)

$a(bx) = (ab)x$

$(a+b)x = ax+bx$

$a(x+y) = ax+ay$

3) $\exists$ an element on R, called 1, s.t. $1x = x$
(multiplicitive identity)

$\exists$ an element, 0, s.t. $0x = 0$

Vector space can be denotes as $\langle S, +, \cdot \rangle$ or S.

A) Examples

1) $\langle \mathbb{R}^3, +, \cdot \rangle$ $\underline{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ $y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$

2) $\langle \mathbb{R}^6, +, \cdot \rangle$ $x = \begin{bmatrix} x_1 \\ \vdots \\ x_6 \end{bmatrix}$ $y = \begin{bmatrix} y_1 \\ \vdots \\ y_6 \end{bmatrix}$

3) S is the set $m \times n$ real matrices

$S = \mathbb{R}^{m \times n}$

4) The set of polynomials with real coefficients with degree $< n$

5) Set of continuous functions

6) $L_p[a,b] = \{$ set of functions in $[a,b]$ s.t.

$$\int_a^b |x(t)|^p \, dt < \infty \, dt \Big\} \quad p \geq 1$$

7) $S = \{(0,0,0,0)\ (0,1,0,1), (1,0,0,1), (1,1,0,0)\}$

$+ :=$ addition % 2 $\qquad R = \{0, 1\}$

Vector space?

Why do we care

The set of vectors = Set of signals